Dependency-Aware Local Typical Sets and a Two-Center $SU(2)$ Atlas

Abstract

We replace a global high-dimensional Gaussian ball by local link, plaquette input, and curvature constraints. The resulting coverage estimate requires no independence. For the finite 2^4 Coulomb Gaussian, the required four-link radius is approximately 0.00602316, reducing the apparent chart gap by a factor of four. We also record an exact two-center atlas for all of $SU(2)$.

1 Local Gaussian coverage

Let A_j be jointly Gaussian blocks in $\mathbb{R}^r$ and suppose their marginal covariance operators satisfy $\mathrm{Cov}(A_j) \preceq \lambda I$. No assumption is made on correlations between different blocks.

Theorem 1 (Local union bound)**.** *For N blocks and $0 < \delta < 1$,*

$$\mathbb{P}\left\{\max_{1\le j\le N} \|A_j\| > \sqrt{\lambda}\left(\sqrt{r} + \sqrt{2\log(N/\delta)}\right)\right\} \le \delta.$$

Proof. Write each marginal as $\Sigma_j^{1/2} Z$ with Z standard Gaussian. Since $\|\Sigma_j^{1/2}\| \le \sqrt{\lambda}$, Gaussian norm concentration gives $\mathbb{P}(\|A_j\| > \sqrt{\lambda}(\sqrt{r} + \sqrt{2x})) \le e^{-x}$. Take $x = \log(N/\delta)$ and apply the union bound. Dependence never enters. □

On the scalar Coulomb space $\ker d_0^T$ of the 2^4 block, the Hessian $d_1^T d_1$ has physical dimension 136 and spectrum in $[2, 12]$. Direct finite spectral calculation gives the leverage coefficients shown below.

Local object	covariance coefficient	radius at $t = 3 \cdot 10^{-6}$
one link ($r = 3$)	0.128788574	0.004025831
four-link plaquette input ($r = 12$)	0.179460637	0.006023158
linear curvature ($r = 3$)	0.685582011	0.009288521

Here $N = 216$ and $\delta = 0.00279270808$. By comparison, the global 408-dimensional ball of radius 0.003 had log mass -803.25266; it was a valid but structurally mismatched certificate.

2 Two-center group atlas

Identify $SU(2)$ with unit quaternions. Choose the nearer center $z \in \{1, -1\}$ according to the sign of the scalar quaternion component and write $zU = \exp X$. Then $\|X\| \le \pi/2$. Thus the two center

charts cover $SU(2)$ with multiplicity at most two. In exponential coordinates the normalized Haar density is

$$J(r) = \frac{1}{2\pi^2}\left(\frac{\sin r}{r}\right)^2,$$

so its ratio on either radius-$\pi/2$ chart is at most $(\pi/2)^2$. Relative tree gauge has constant gauge-volume Jacobian.

3 Scope

The local-union theorem and group atlas do not yet compare normalized nonlinear and Gaussian conditional measures. A charged estimate for the nonprincipal center and orbit channels is also absent. The next question is whether nonlinear quotient convexity reaches the radius 0.006023158.